



《CAD与工程制图》

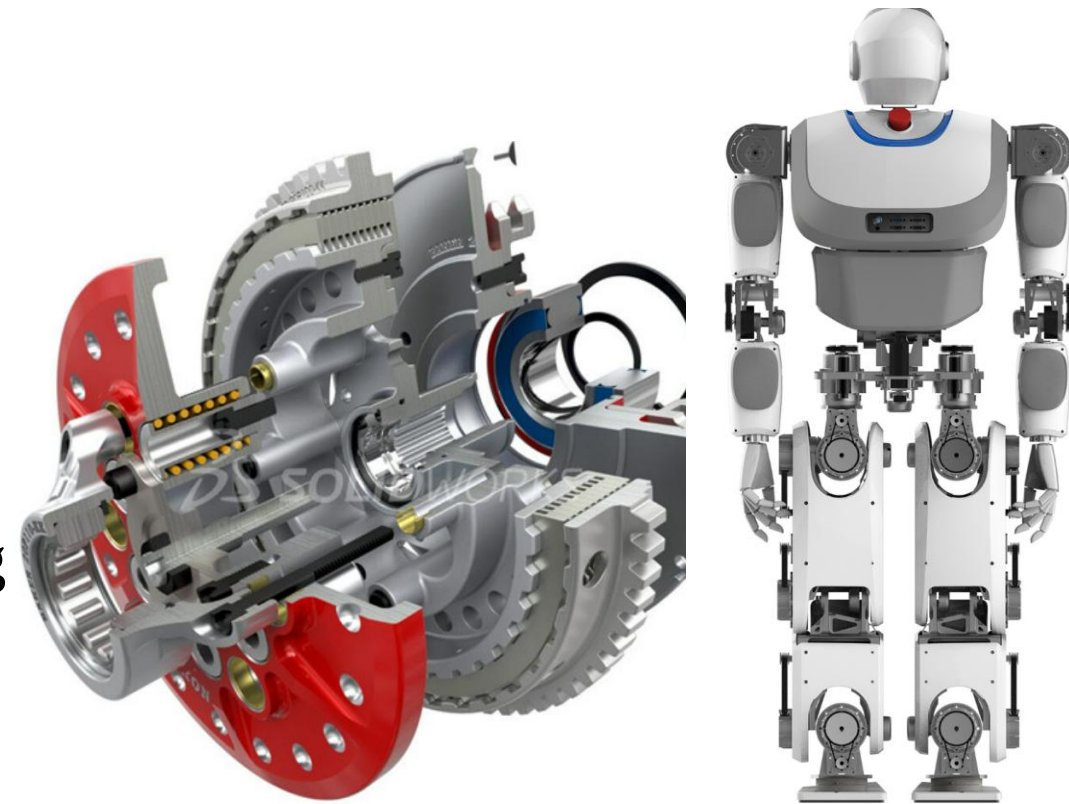
CAD (Computer Aided Design) and Engineering Drawing

2025年 春季

Yixuan Guo 郭艺璇

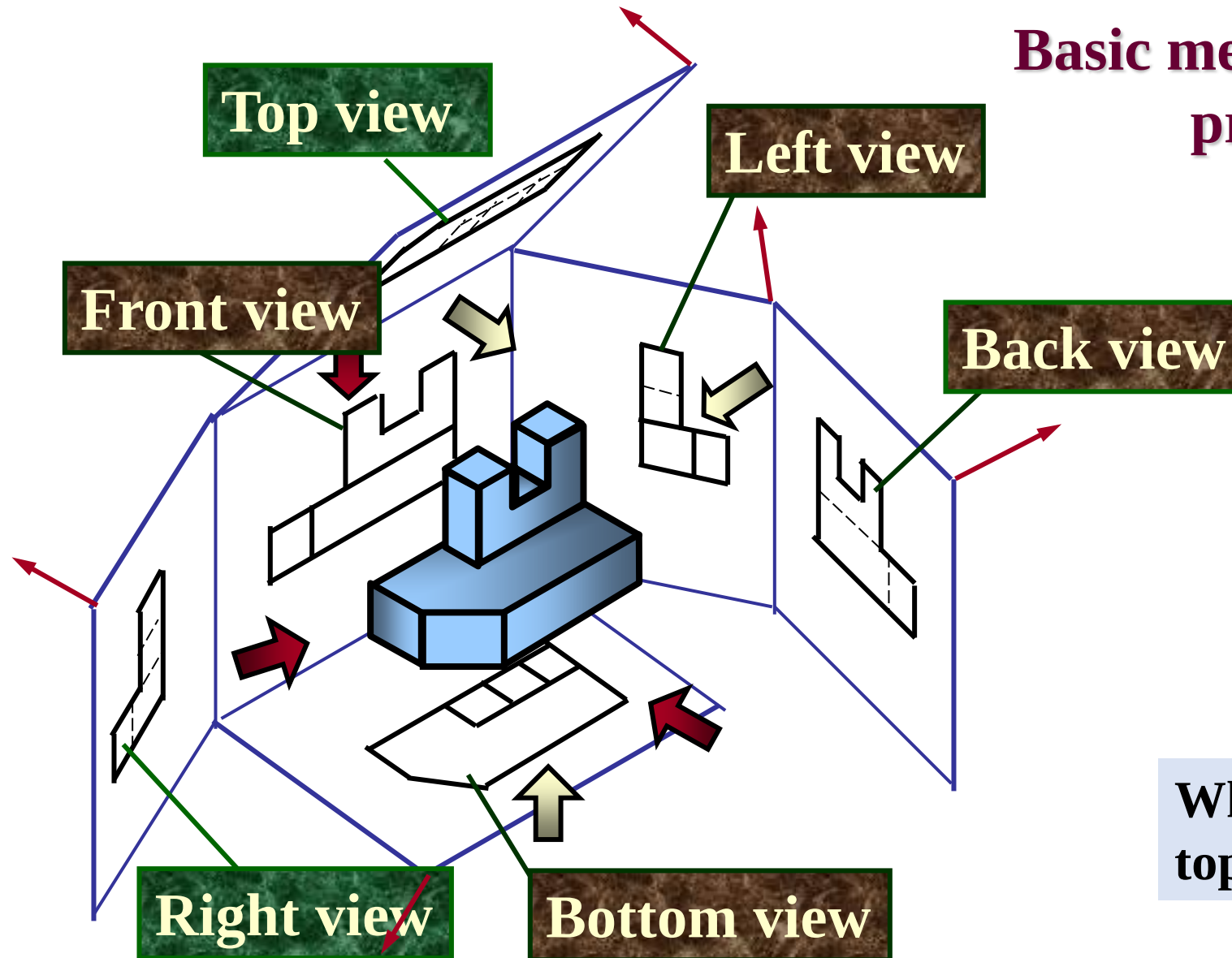
Department of Mechanical and Energy Engineering

Southern University of Science and Technology



Chapter 6 Common Methods of Part Representation

- **6-1 Views**
- **6-2 Section Views**
- **6-3 Cross-Sectional Views**
- **6-4 Standard and Simplified Drawing Methods**



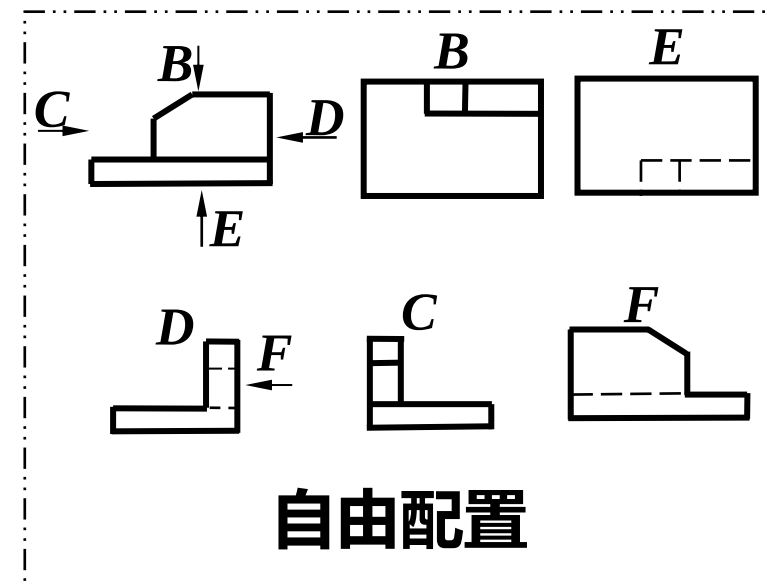
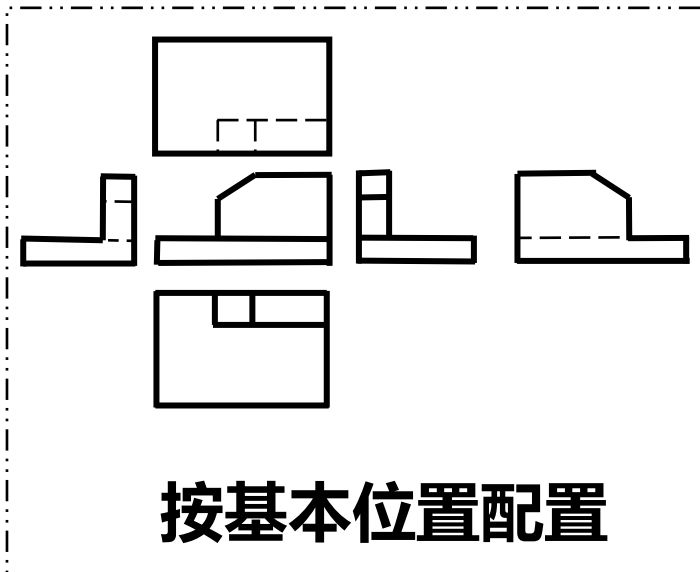
Basic methods of unfolding the projection planes

**Measuring corresponding relationships:
Still follow the "three equals" rule.**

When are the right, back, and top views used?

6.1.1 Auxiliary views

Auxiliary views are **freely positioned** views

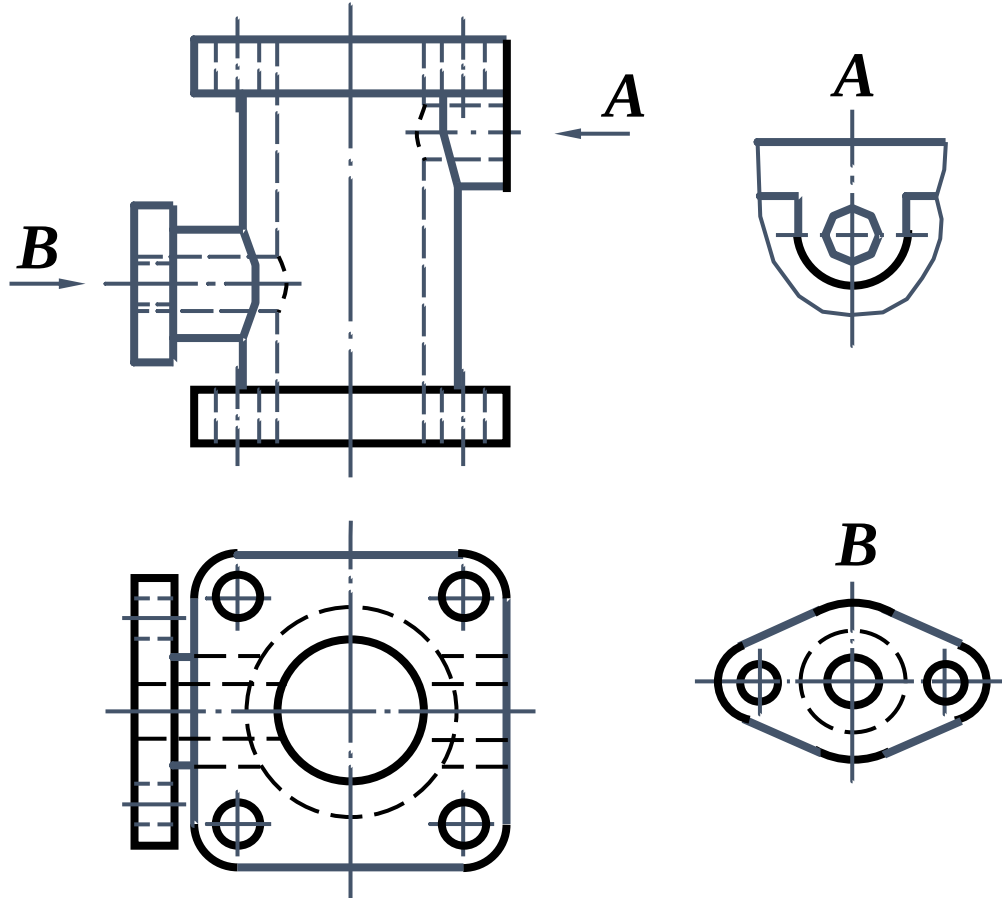


※ Label **auxiliary views with letters above them**. Use **arrows to show projection direction** near views, matching with letters.

※ Preferably place arrow on front view, **except for rear view**.

6.1.2 Local views

Local views are views obtained by **projecting a part of an object** onto a primary projection plane.

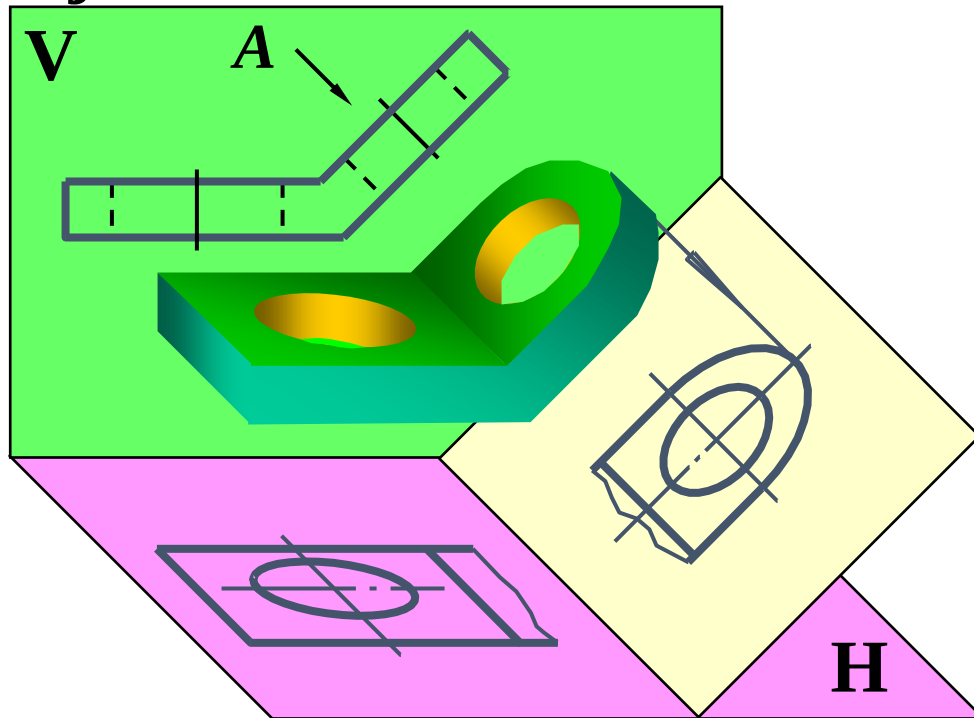


Notes:

- Use **arrows labeled with letters** to indicate the **section and projection direction**, and label the view name.
- The **scope** of local views is indicated with **dashed lines**. When the local structure is complete and the outer contour is closed, dashed lines can be **omitted**.
- Local views can be arranged in the same way as primary views or auxiliary views.

6.1.3 Oblique views

Issue: When an object's surface is inclined relative to the projection plane, its projection does not reflect the true shape.



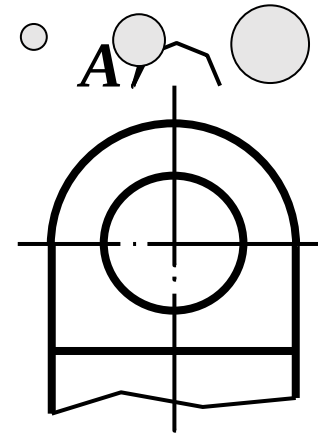
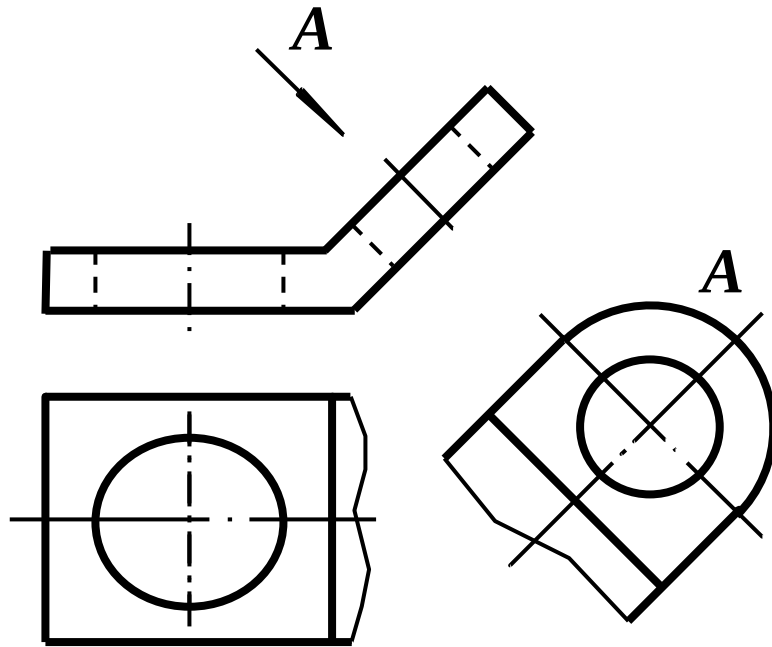
Solution:

- ★ Add an auxiliary projection plane parallel to the inclined surface.
- ★ Project the inclined part onto the auxiliary projection plane.

Oblique view is a view obtained by projecting an object onto a plane **not parallel to the primary projection planes**.

6.1.3 Oblique views

Drawing Method

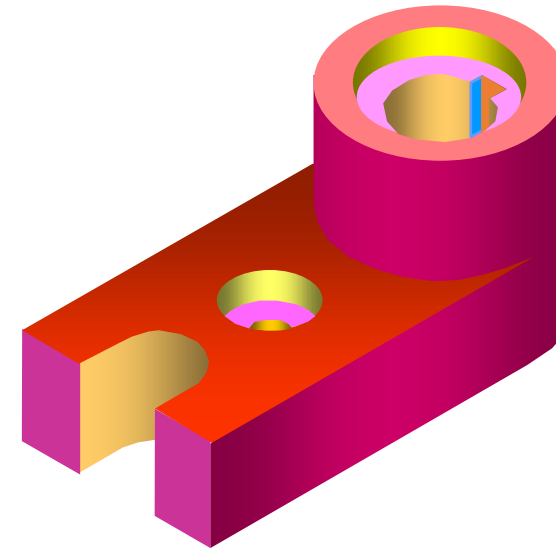
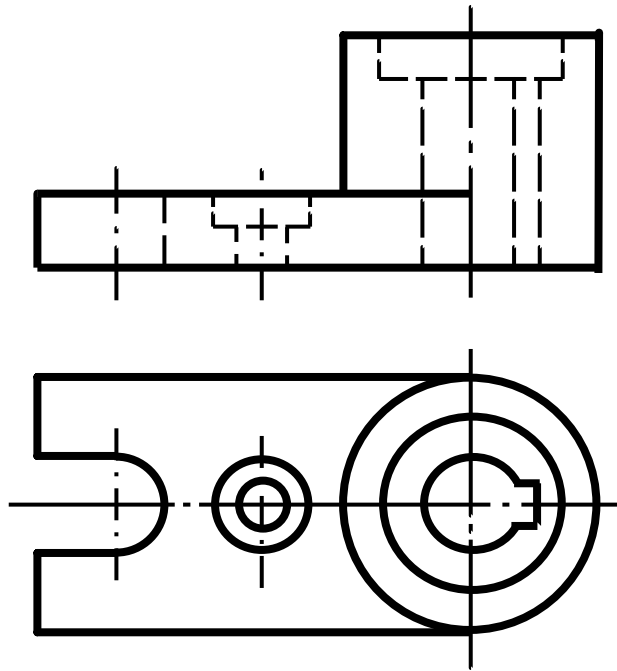


Letter near arrow tip,
symbol direction
consistent with
rotation direction.

Notes:

- The **break boundary** of oblique view is indicated with **dashed or double broken lines**.
- Oblique views are typically placed and labeled according to projection direction.
- **Rotation** of oblique view is allowed, but the **rotation direction** must be noted above the oblique view.

Issue: When the internal **shape of a part is complex**, many dashed lines will appear on the views, making it difficult to read the drawing and dimension it. How should the view be expressed in such cases?



Solution?

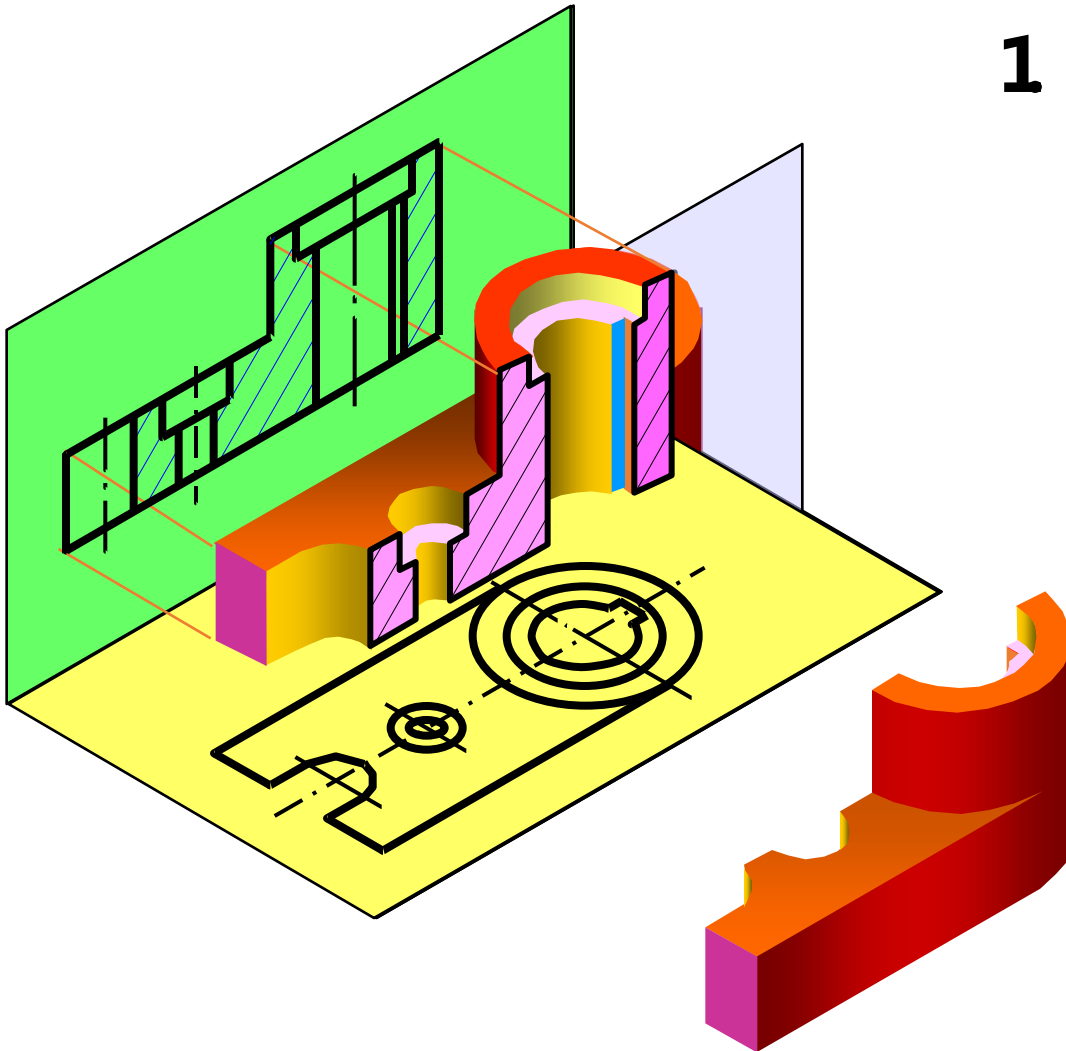
Use section views

Chapter 6 Common Methods of Part Representation

- **6-1 Views**
- **6-2 Section Views**
- **6-3 Cross-Sectional Views**
- **6-4 Standard and Simplified Drawing Methods**

6.2.1 Concept of Section Views

1、 Formation of Section Views



- Imagine **cutting** the part with a **cutting plane**.
- **Remove** the obstructing part.
- **Project** the sectioned area **onto** the **projection plane**.
- **Add section lines** **within** the **sectioned area**.

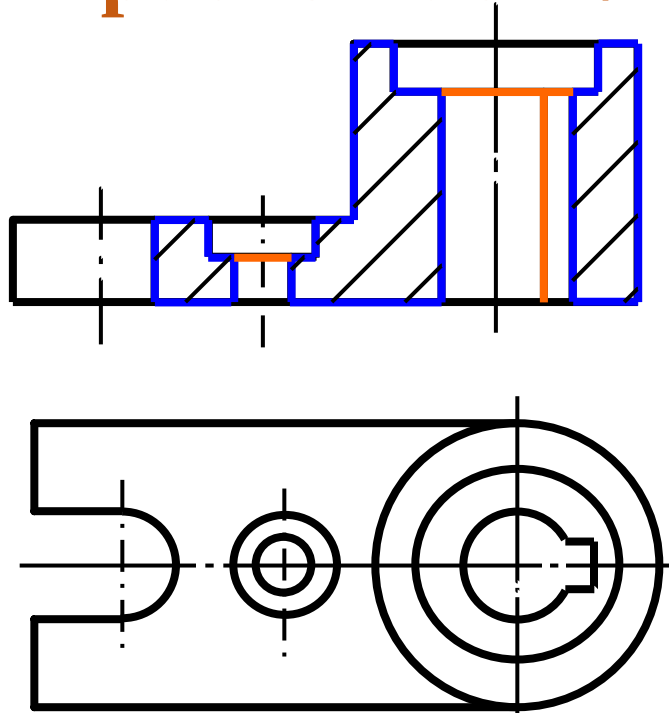
6-2 Section Views



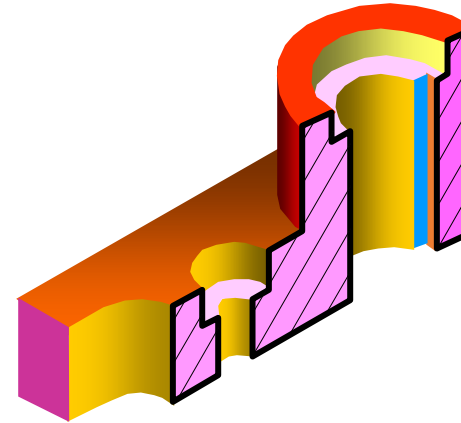
Section symbols for various materials

金属材料 (Metal Materials)		Wooden plywood	
Winding Components		Soil	
Stacked Sheets of Cores, Coils, Transformers, and Reactors		Concrete	
Plastic, Rubber, Oilcloth, and other Non-metal Materials		Reinforced Concrete	
Molding sand, filled sand, ceramic blades, etc.		Brick	
Glass and Other Transparent Materials		Mesh	
木材 Wood	纵断面 横断面 	Liquid	

6.2.1 Concept of Section Views



2、 Drawing Steps



- ❖ 1. Determine the **position of the cutting plane** — parallel to the projection plane, symmetrical plane, or axis.
- ❖ 2. **Draw the sectioned part and the visible parts after cutting** — think which parts are removed? the shape of the sectioned area? and which parts are visible during projection?
- ❖ 3. Add **section symbols** within the sectioned area.

6.2.1 Concept of Section Views

3. Labeling of Section Views

① Cutting Line

Indicates the position of the cutting plane (thin line with single dots). Generally, it can be omitted.

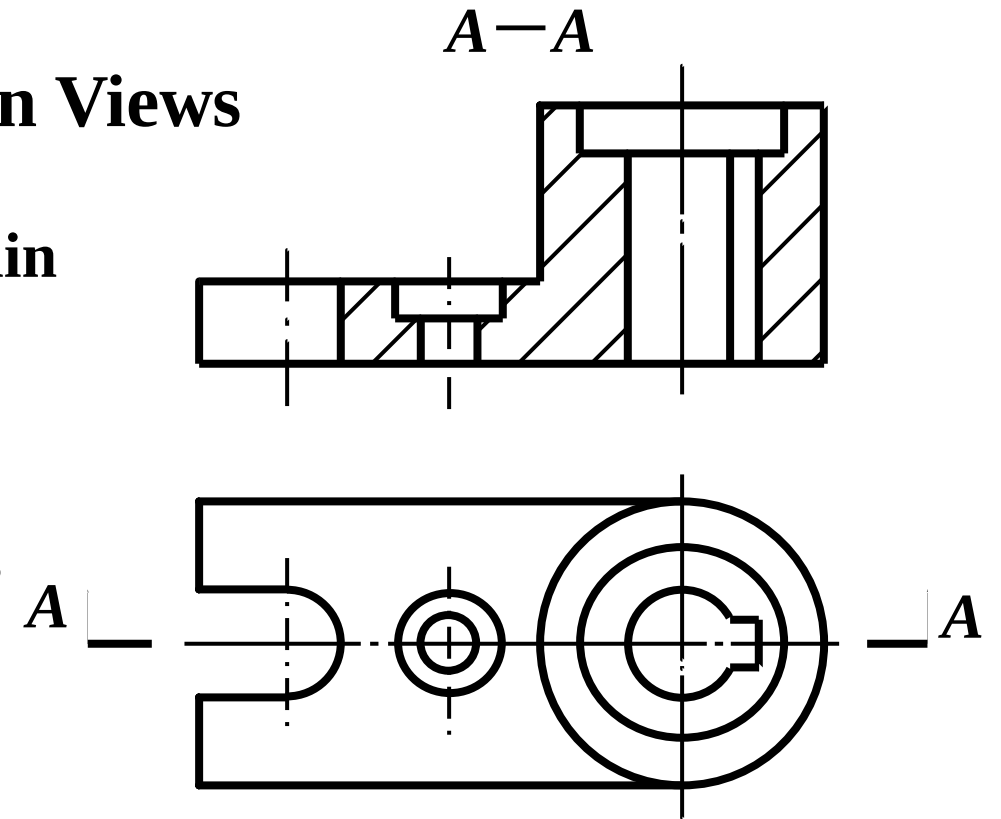
② Cutting Symbol

Represents the **start, end, and turning** points of the cutting plane and its **projection direction**.

③ Name of Section View, e.g., X-X

The following cases can omit labeling:

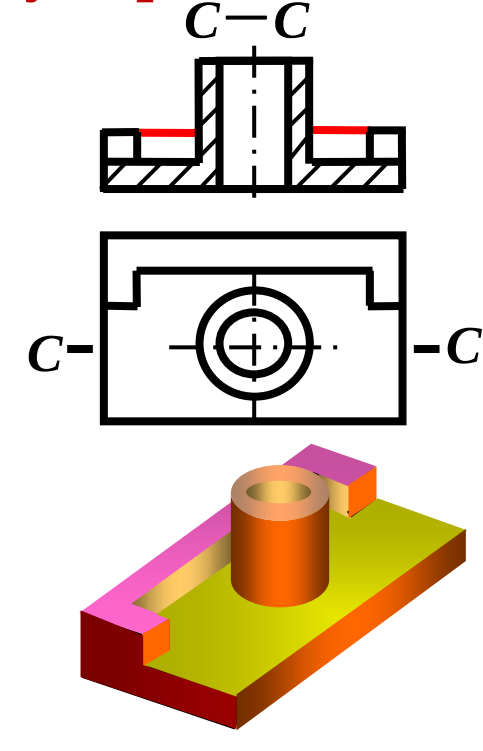
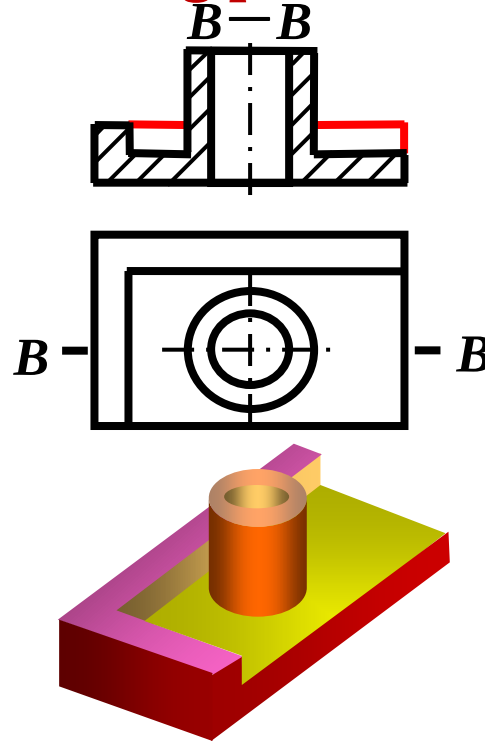
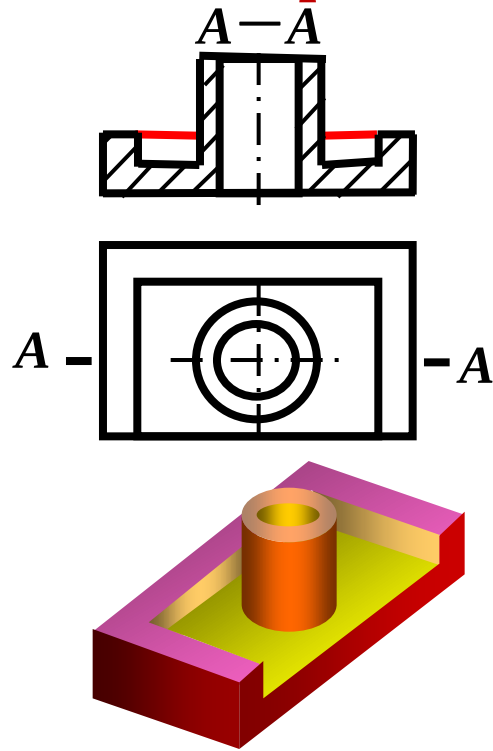
- ◆ When **the section view is arranged according to the basic view** relationship, the arrow can be omitted.
- ◆ When a single cutting plane passes through the **symmetrical (or essentially symmetrical) plane** of the part, **the section view is arranged according to the basic view**.



6.2.1 Concept of Section Views

4. Notes and Considerations

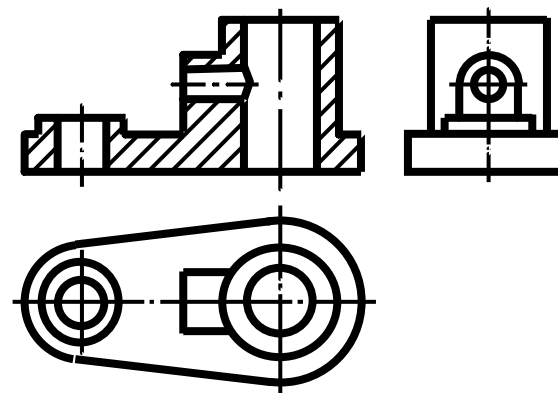
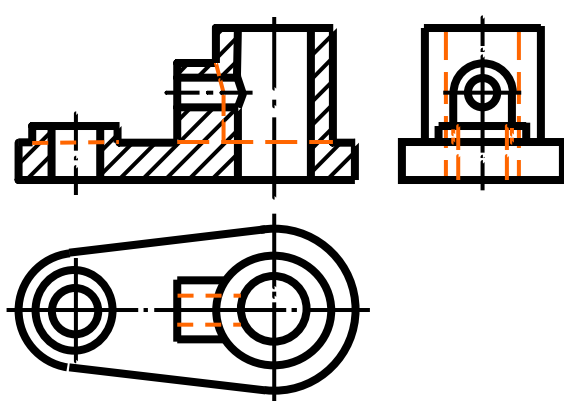
- ① Selection of the cutting plane: through the part's **symmetry plane, axis, or parallel to the projection plane.**
- ② Cutting is **hypothetical**; other views should still be drawn completely.
- ③ **All visible parts behind the cutting plane must be fully depicted.**



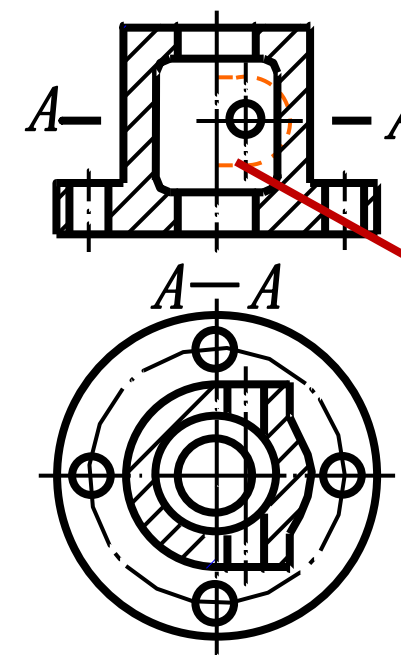
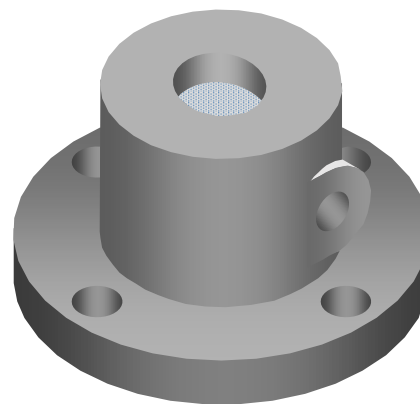
6-2 剖视图



④ For structures that have been clearly expressed in a sectional view, if the projection of this part of the structure in other views is a dashed line, the dashed line is omitted and not drawn.



However, the clear structure was not indicated, and a few dotted lines were allowed to be drawn.

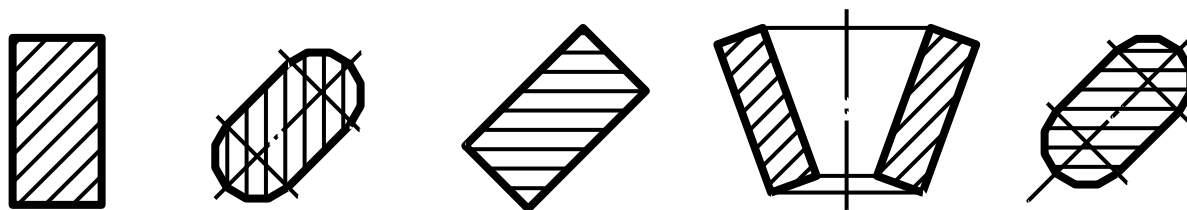


The dotted line of the main view boss needs to be drawn to express the structure.

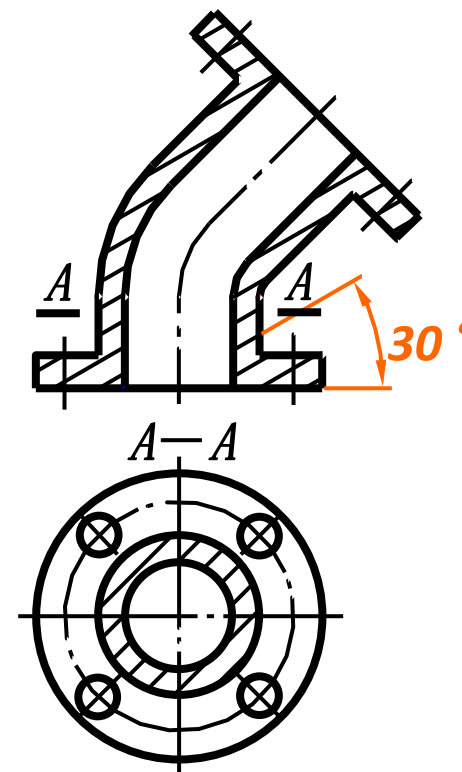
6-2 剖视图



⑤ When it is not necessary to indicate the category of the material in the section area, the section symbol can be represented by a universal section line. The general section line is a thin solid line, preferably forming a 45° Angle with the main contour of the figure or the symmetry line of the section area. For each cross-sectional area of the same object, the drawing method of the cross-sectional line should be consistent.



When the drawn profile line is parallel to the main contour line of the figure or the axis of the profile area, the profile line of the figure should be drawn at an Angle of 30° or 60° to the horizontal.



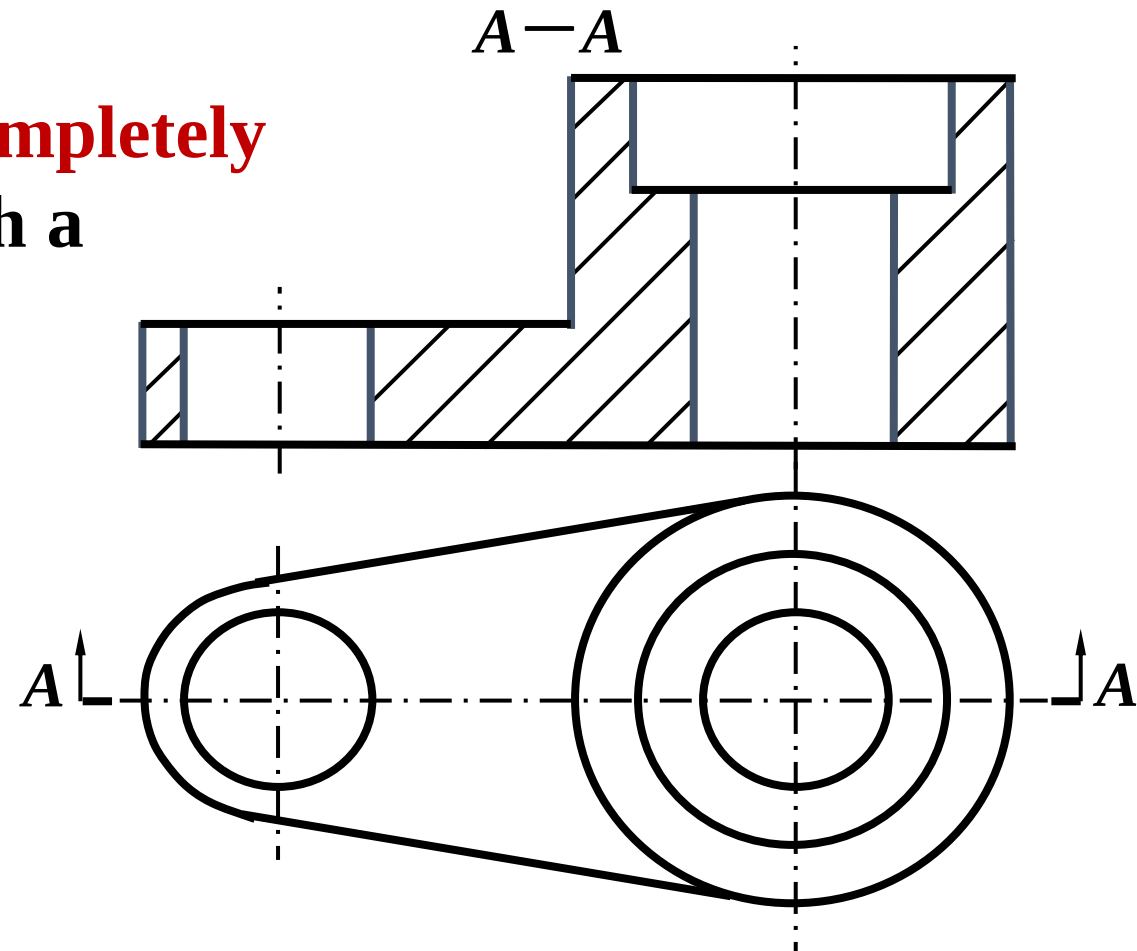
6.2.2 Types of Section Views

1. Full Section View

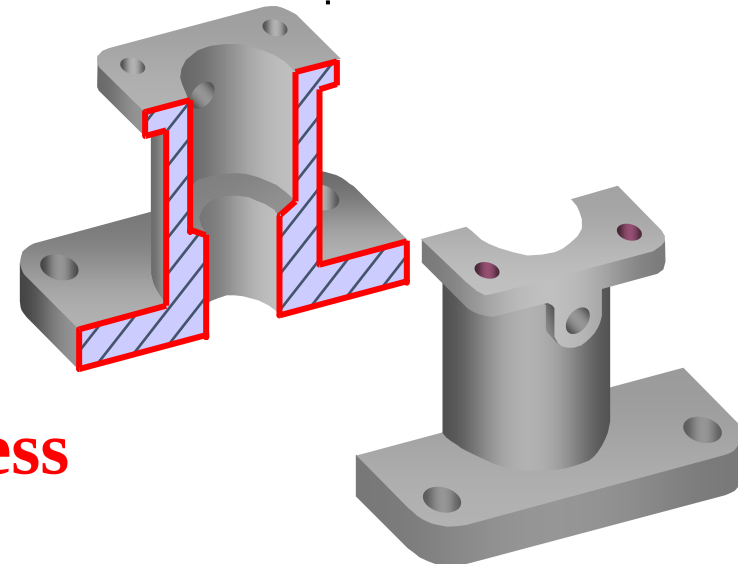
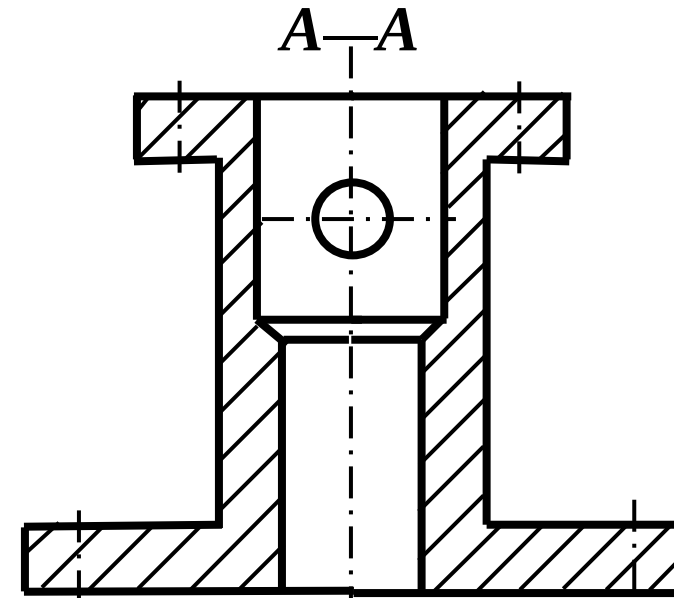
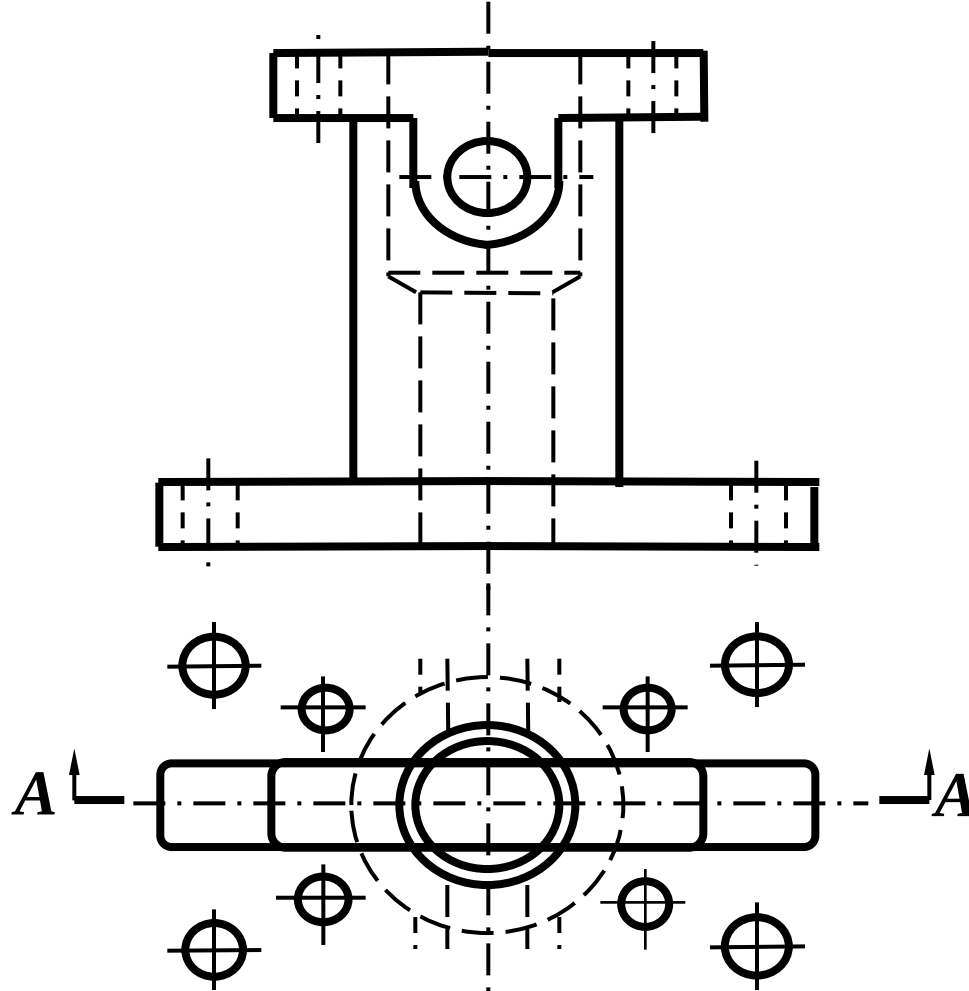
A section view obtained by **completely cutting through** an object with a cutting plane.

Applicable to parts with:

- Simple exterior shapes.
- Complex and asymmetrical internal structures.



2. Half Section View

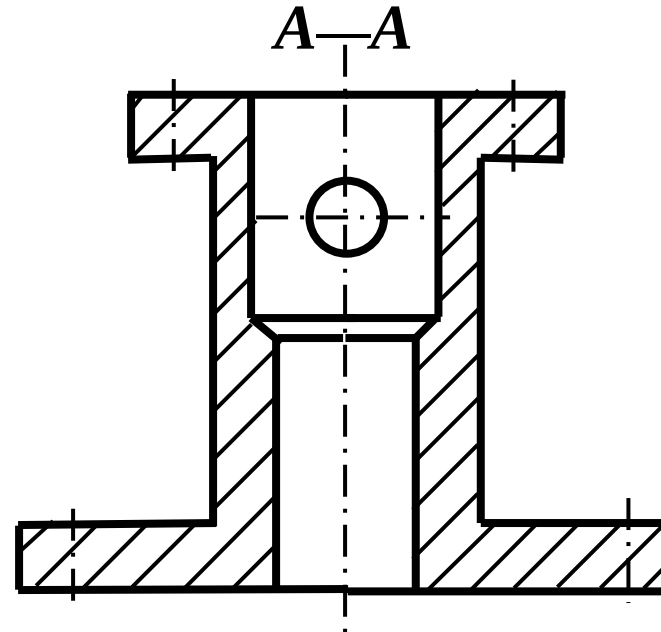
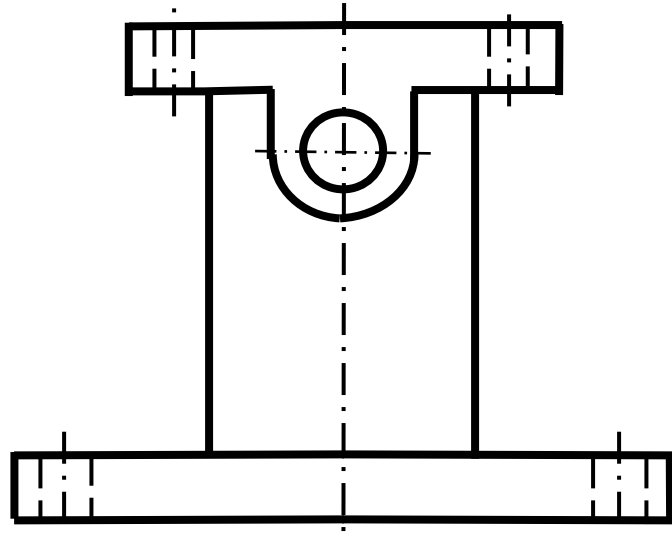


When symmetry, a full section cannot express the exterior shape, what to do?

6-2 Section Views



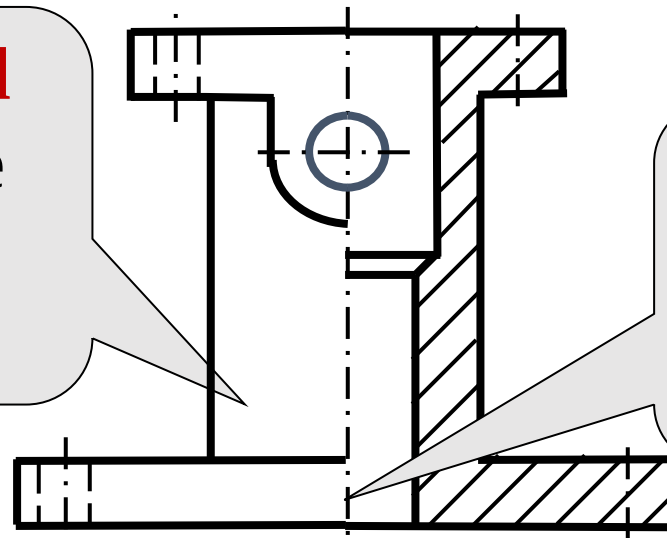
Solution:



Half Section :

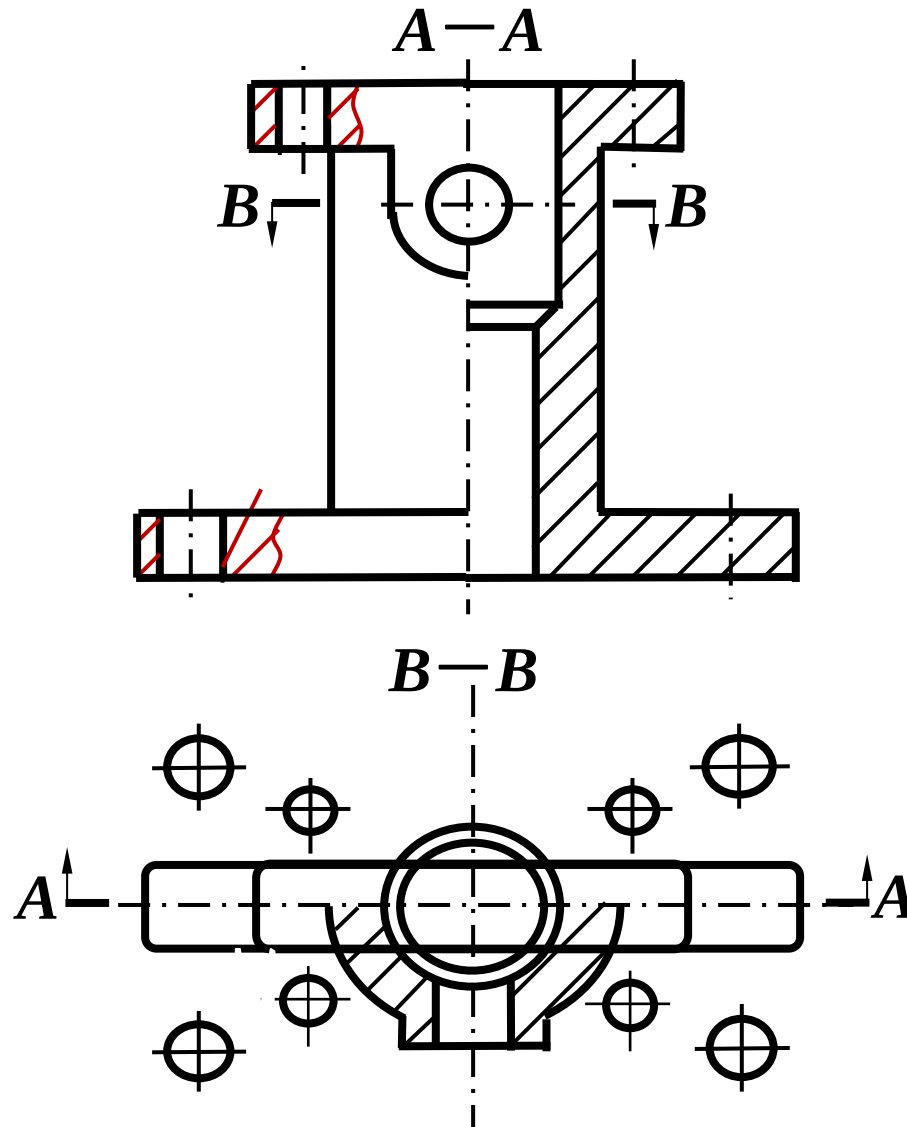
Divide along the line of symmetry, draw one half as a view and the other half as a section.

Do not draw internal dashed lines that are already clearly expressed.



The boundary between the view and the section view should be a **thin dotted line**, not a thick solid line.

3. Local Section View



A section view obtained by **cutting through a part of an object** with a cutting plane.

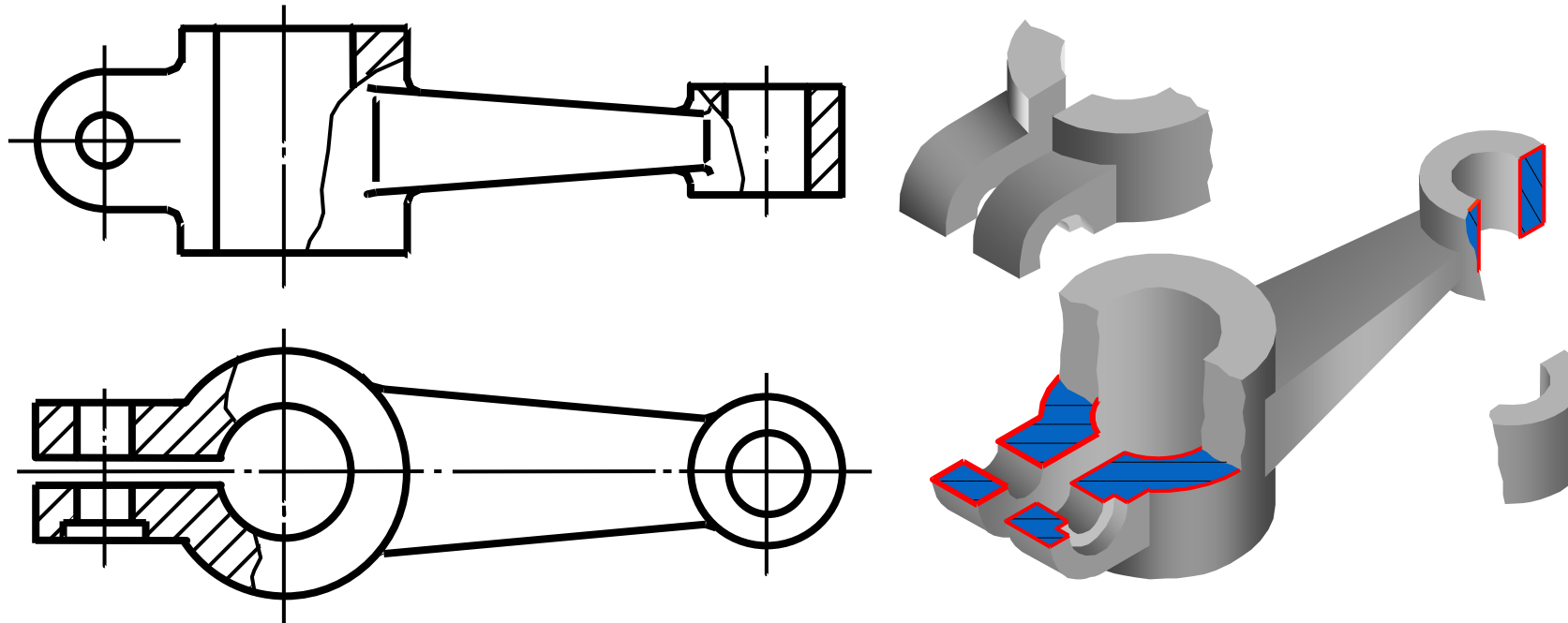
6-2 Section Views



Applicable to parts with:

Local section is a flexible representation method with a wide range of applications:

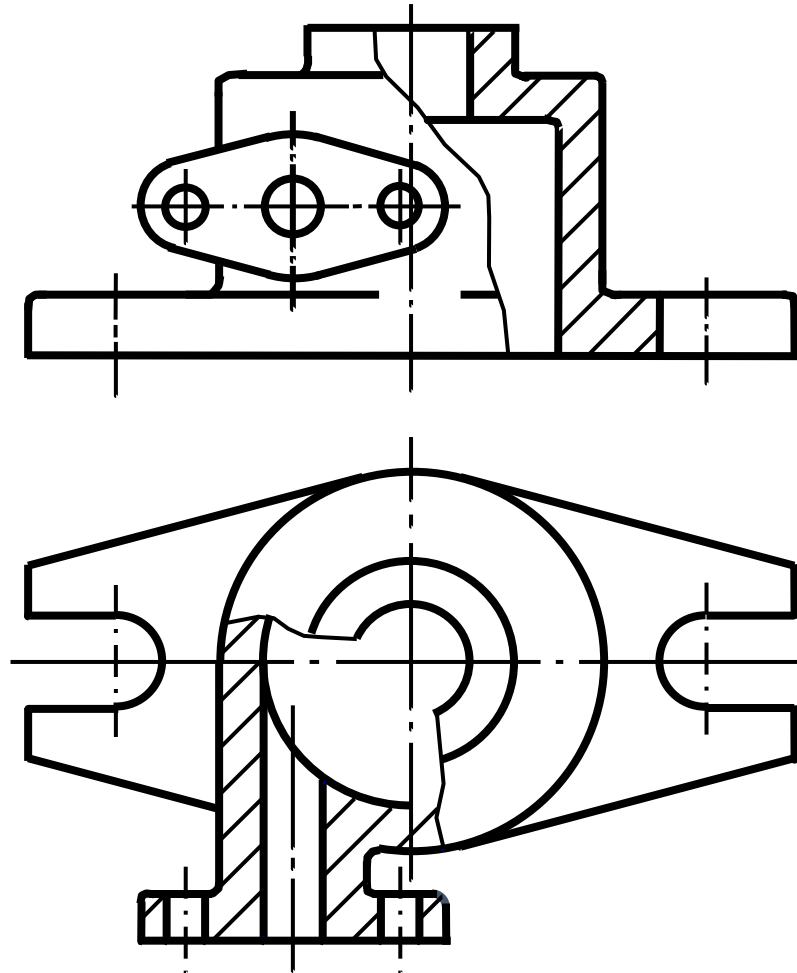
① When only a local internal shape **needs to be sectioned** and it is **not suitable for a full section**.



6-2 Section Views



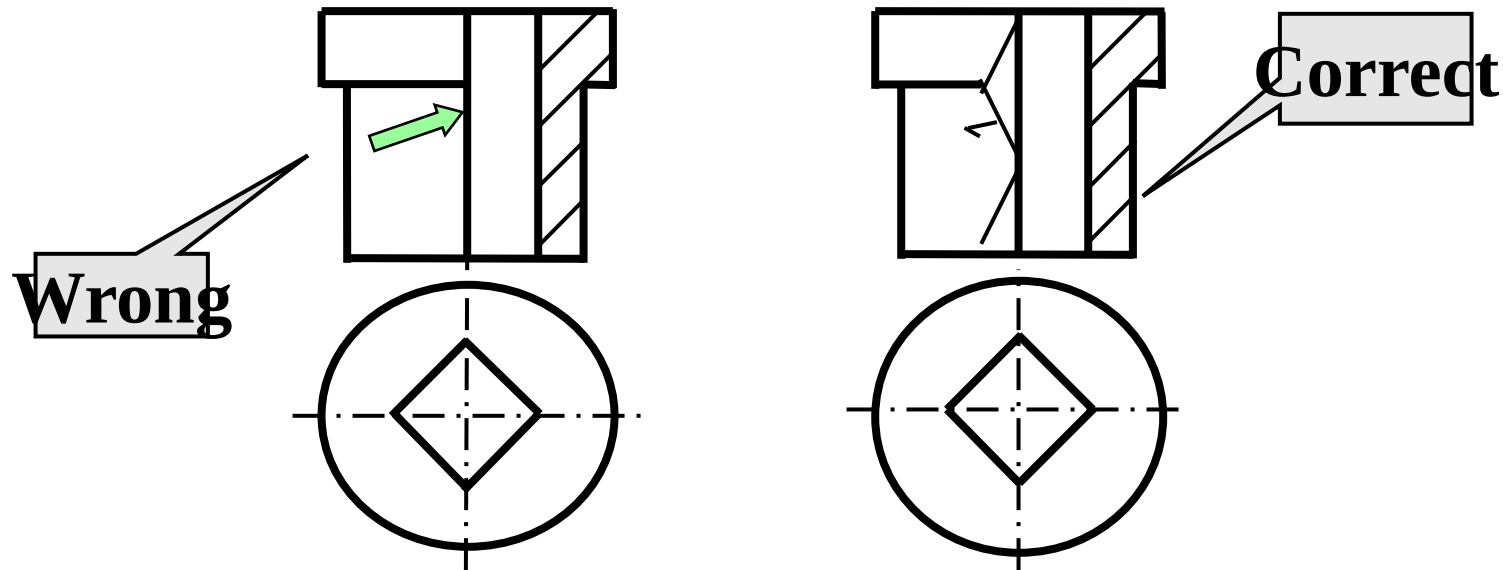
② When **both the internal and external shapes** of an **asymmetric** part need to be expressed.



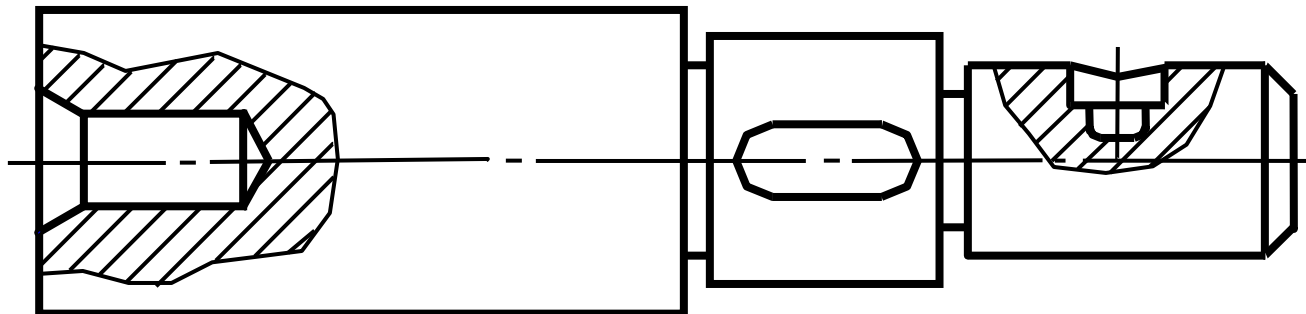
6-2 Section Views



③ For symmetric parts **where the contour lines coincide with the center lines**, making a half section view unsuitable; a local section view can be used.



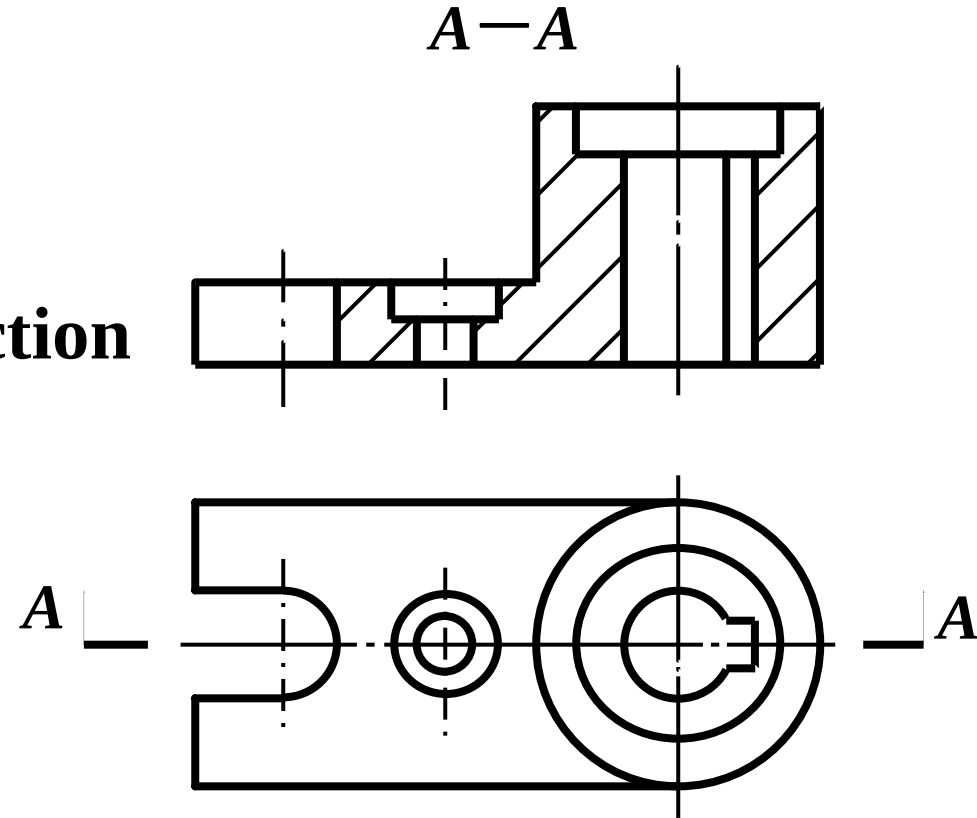
④ When there are **holes or slots** on a solid rod, a local section view should be used.



6.2.3 Types of Cutting Planes

1. Single Cutting Plane

(1) Parallel to a Basic Projection Plane



1. Single Cutting Plane

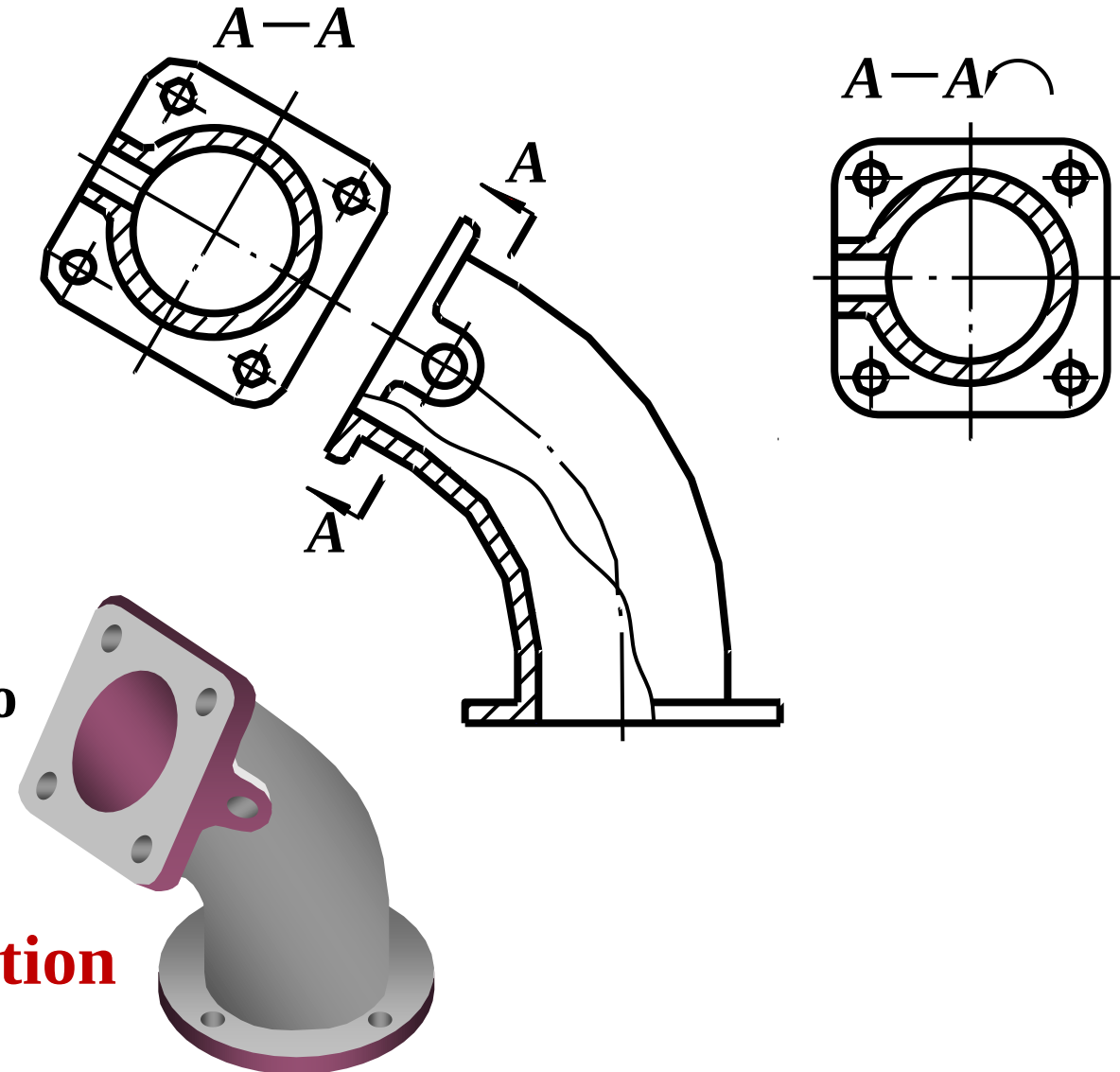
(2) Not parallel to any basic projection plane

★ Labeling Method:

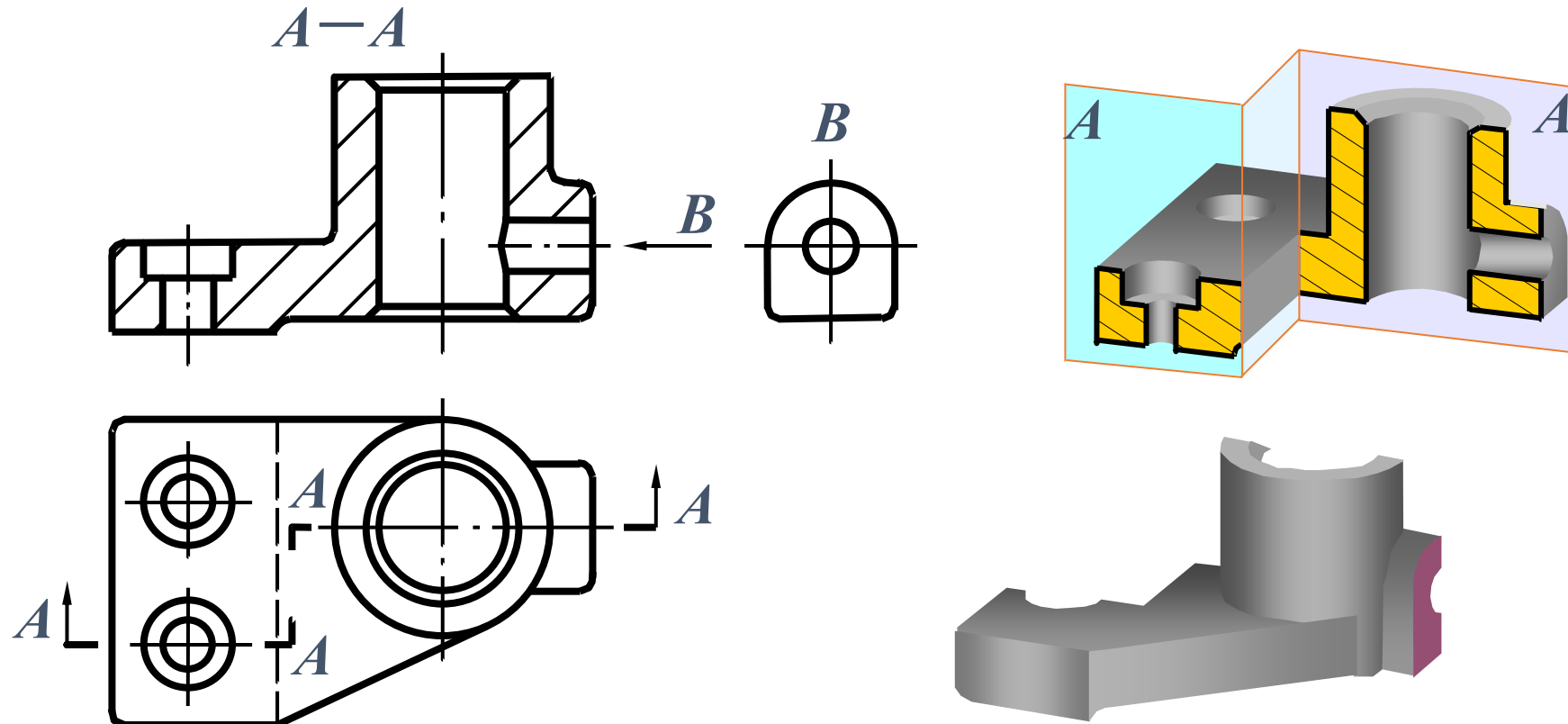
★ Applicable Scope:

When a part has **an inclined section** where **both the internal and external** shapes need to be depicted.

★ This section view can be arranged according to the **oblique view configuration method**.



2. A Set of **Mutually Parallel** Cutting Planes



☆ **Labeling Method:**

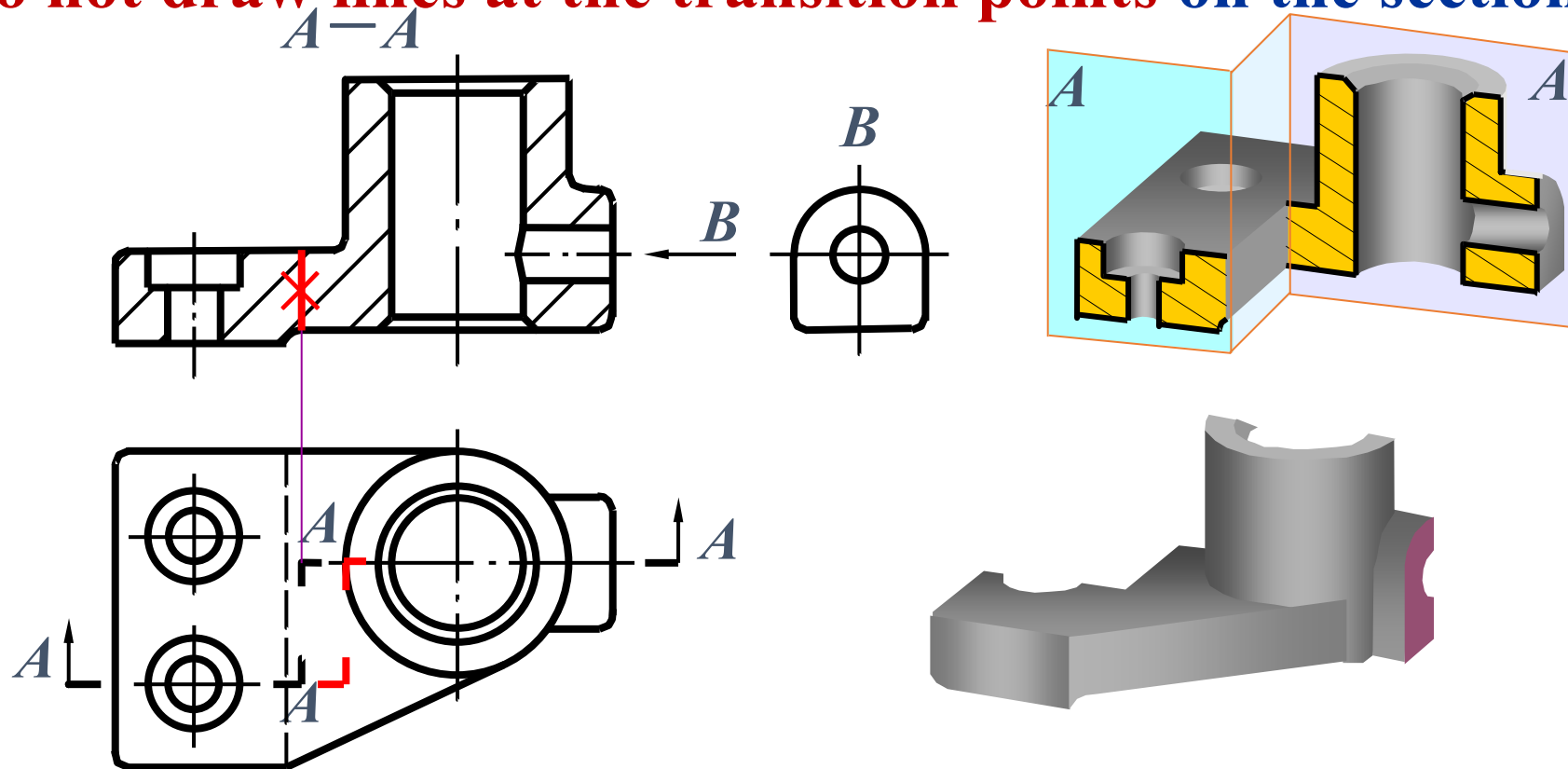
☆ **Applicable Scope:**

When the internal structures such as **holes, slots, and cavities** of a part are **not on the same plane**.

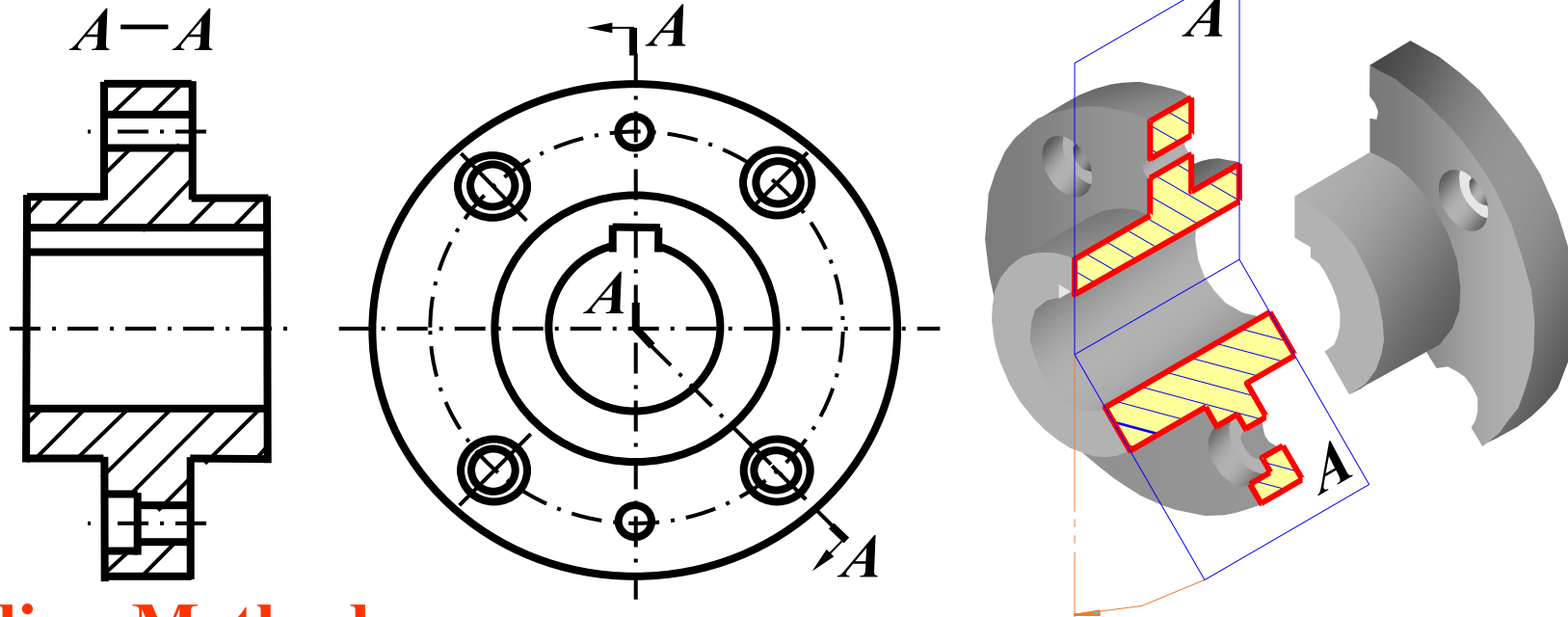
2. A Set of Mutually Parallel Cutting Planes

★ **Issues to Note::**

- ① **The transition between two cutting planes should not coincide with the contour lines on the drawing.**
- ② **Do not draw lines at the transition points on the section view.**



2. A Set of Mutually Parallel Cutting Planes



- ★ Labeling Method:
- ★ Applicable Scope:

When the internal structure of a part **cannot be fully expressed with a single cutting plane**, and the part **has a rotational axis**.

Practice problems

**Let's practice together, and I'll make sure
you understand!😊**

Chapter 6 Common Methods of Part Representation

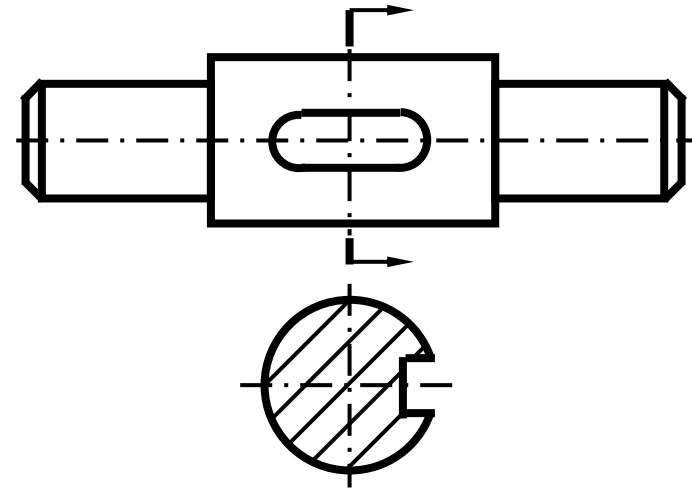
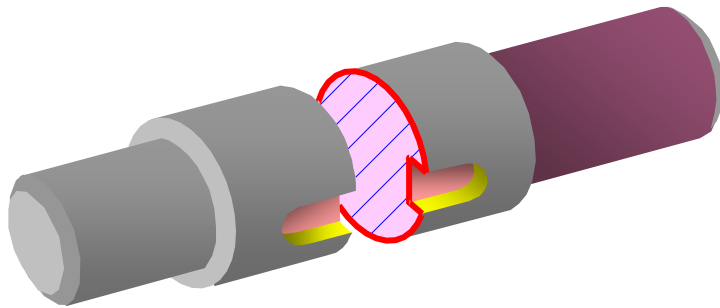
- **6-1 Views**
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- **6-3 Cross-Sectional Views**
- **6-4 Standard and Simplified Drawing Methods**

6-3 Cross-Sectional Views



6.3.1 Concept of Cross-Sectional Views

Imagine **cutting an object** at a certain point with a cutting plane, and then drawing only the part of the object **in contact with the cutting plane (the cross-sectional area)**.



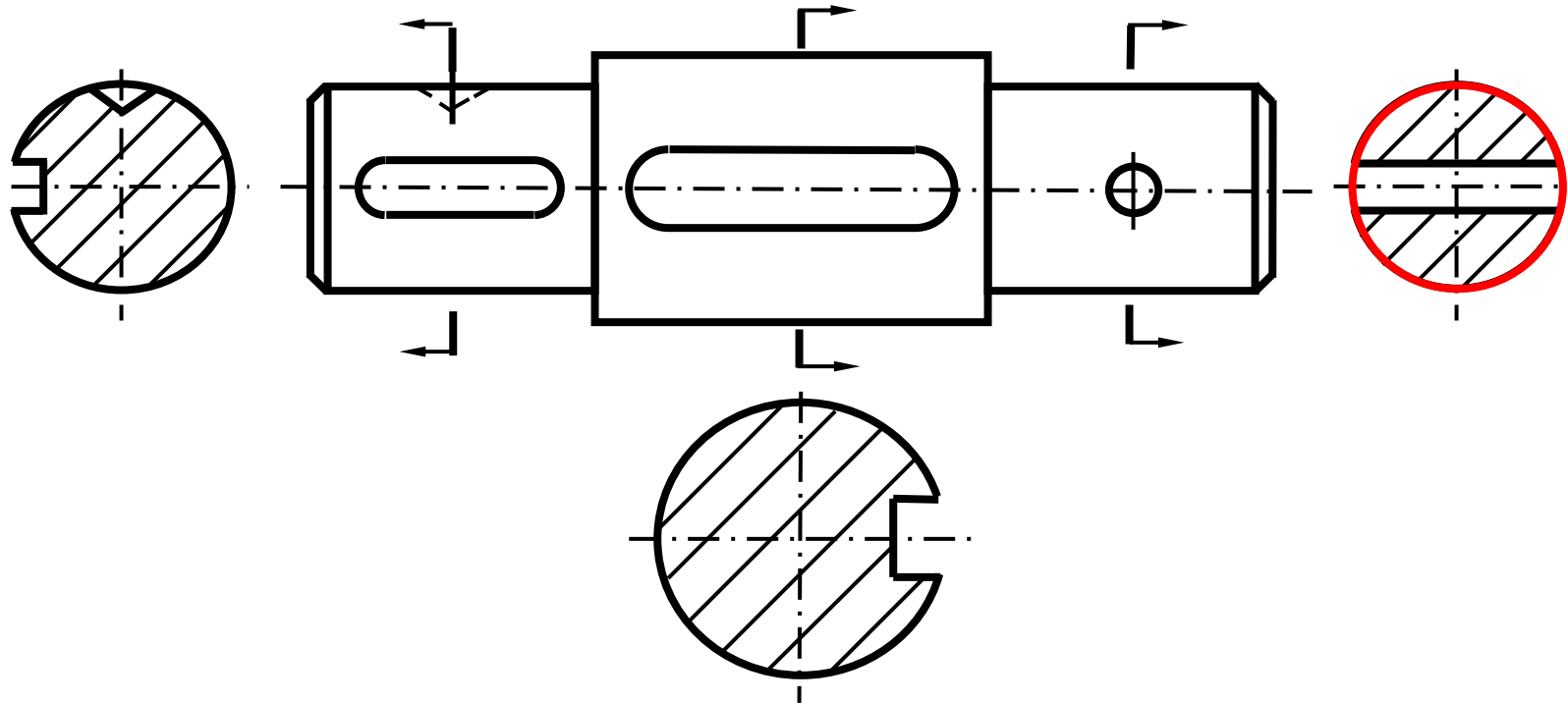
6.3.2 Types of Cross-Sectional Views

1. Removed Section View

(1) Method

Drawn outside the view, with the outline in thick solid lines.

Placed on the extension of the cutting line or other suitable positions.



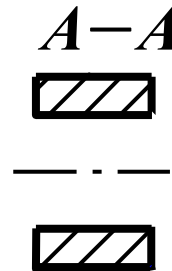
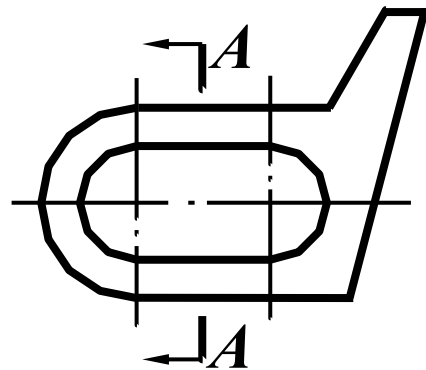
Notes:

☆ When the cutting plane passes through the axis of a hole or depression formed by a surface of revolution, it should be drawn as a section view.

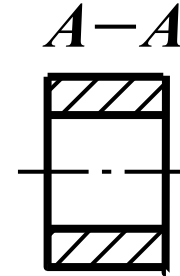
6-3 Cross-Sectional Views



★ When the cutting plane passes through a non-circular hole, resulting in **two completely separated cross-sections**, these structures should also be drawn as section views.



Wrong

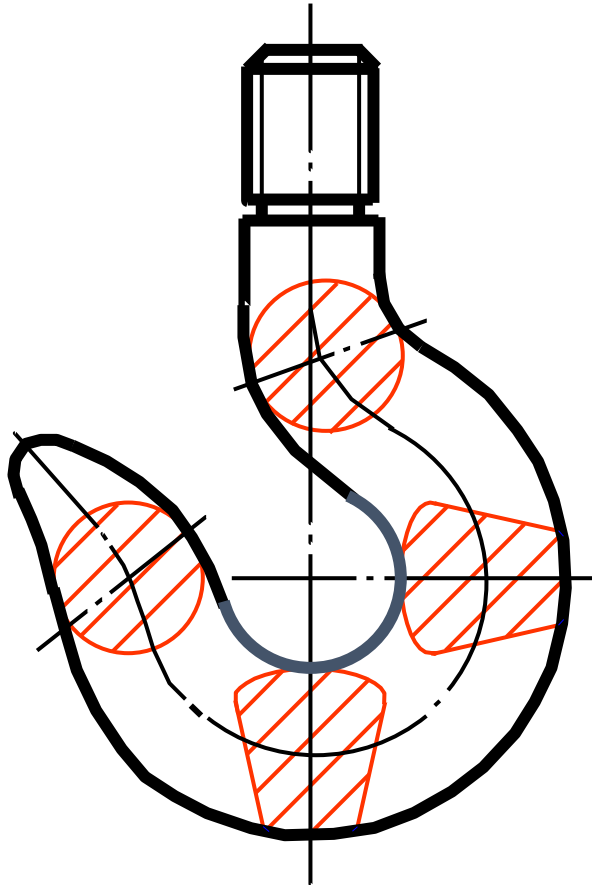


Correct

6.3.2 Types of Cross-Sectional Views

2. Coinciding Section View

(1) Method

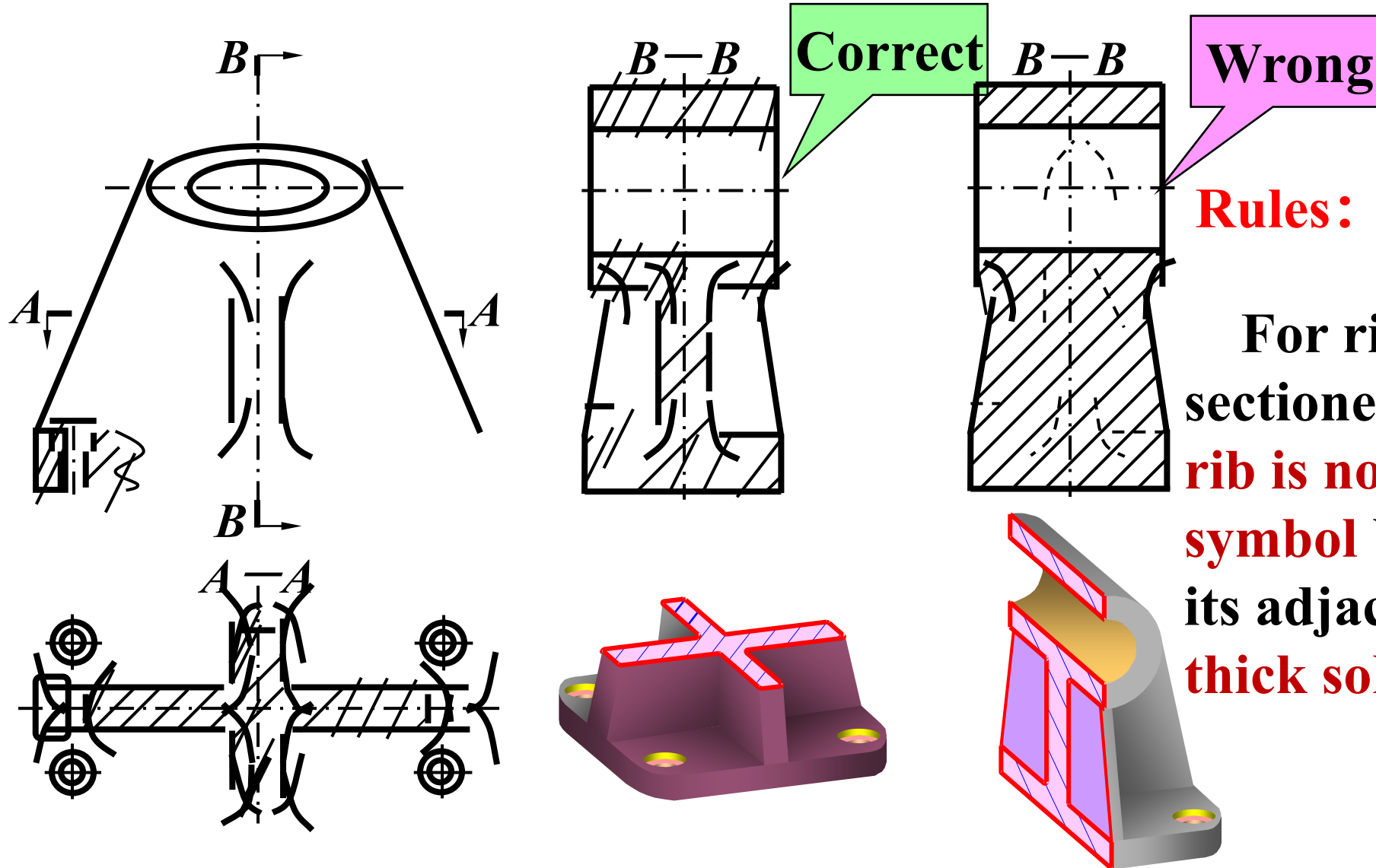


Drawn within the view, with the outline in thin solid lines.

Chapter 6 Common Methods of Part Representation

- **6-1 Views**
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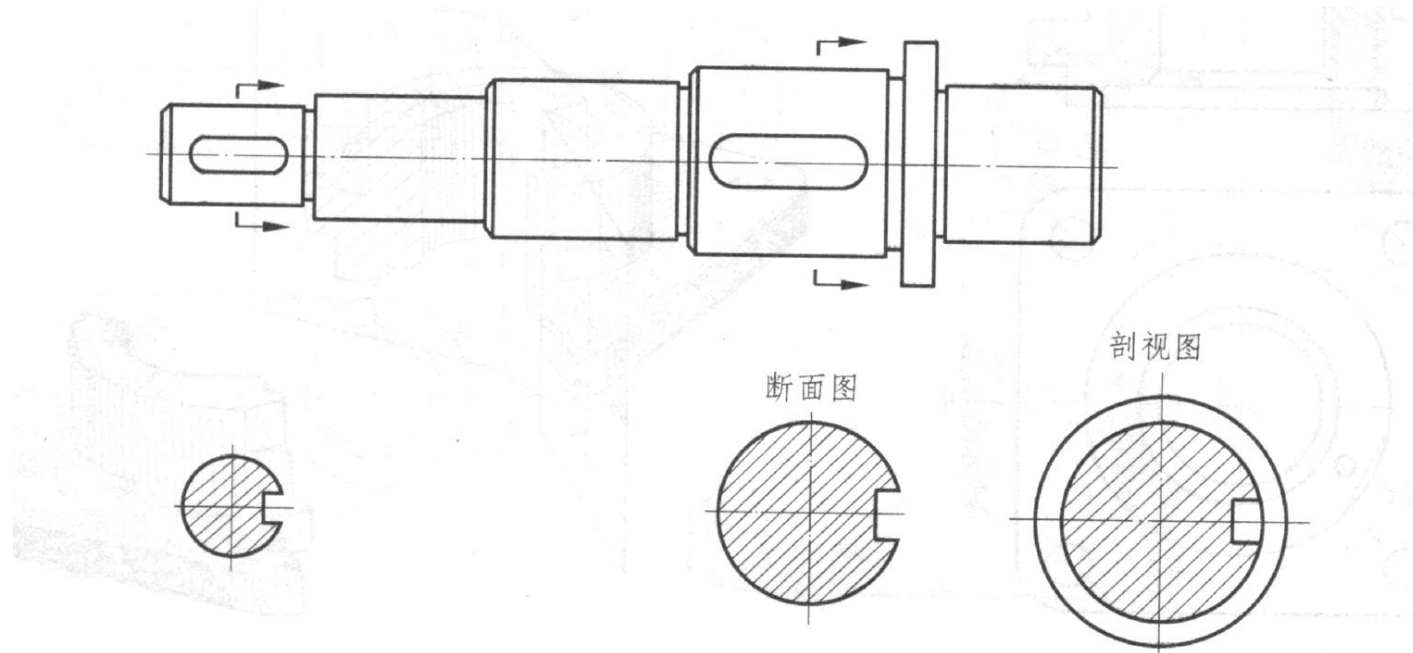
6.4.1 Drawing Method for Rib Plates



Differences between Cross-Sectional Views and Section Views

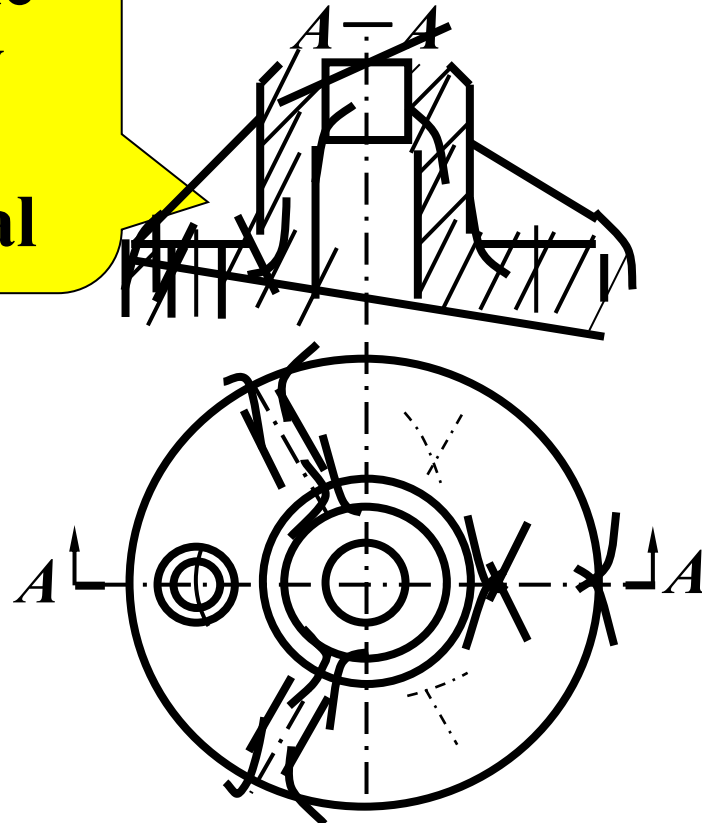
Cross-Sectional Views: **Only the shape of the sectioned surface** of the part is shown.

Section Views: In addition to showing **the shape of the sectioned surface**, the **projection of the remaining part** after sectioning is also depicted.

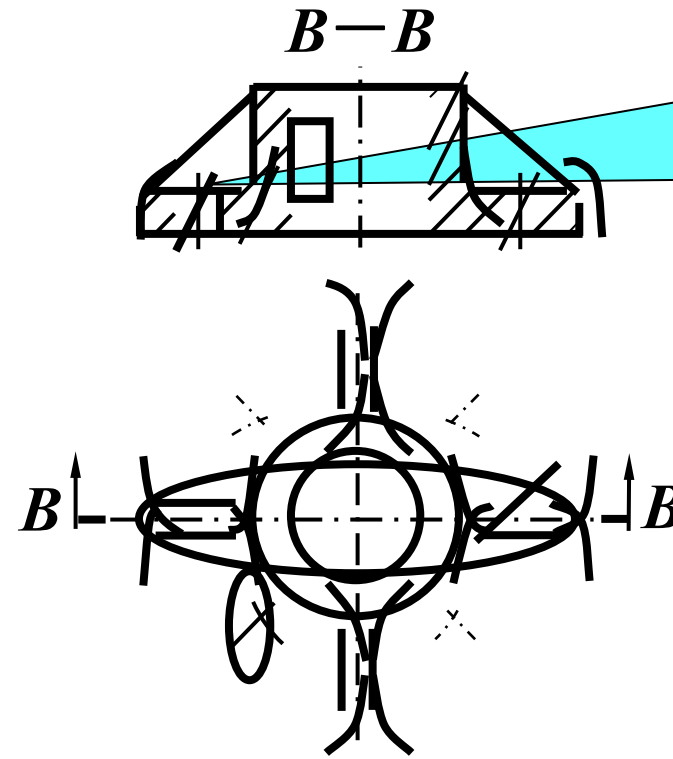


6.4.2 Drawing Method for Uniformly Distributed Ribs and Holes

Asymmetric
ribs, draw
them
symmetrical



Unsectioned
holes, draw
them sectioned

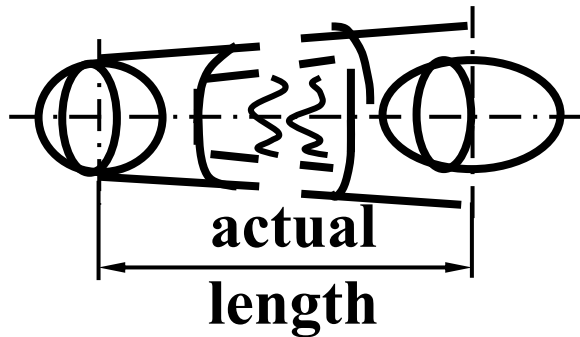


For several **holes** of the **same diameter** and **regularly distributed**, only one or a few need to be drawn; the rest can be indicated by center lines with **fine dotted lines**.

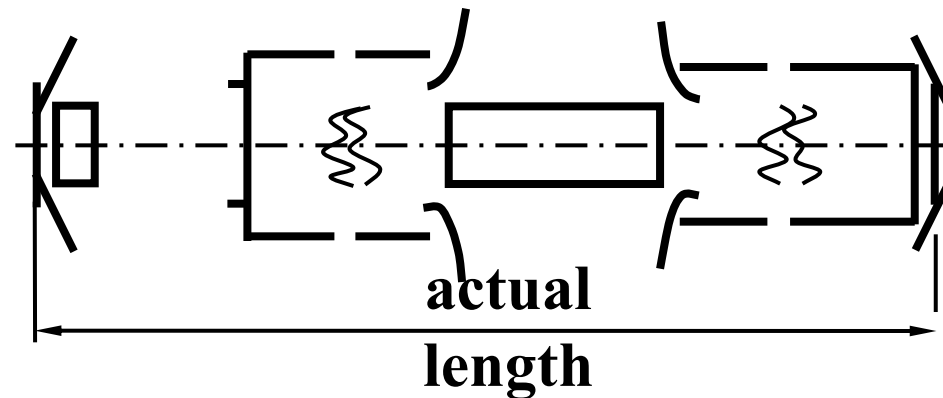
6.4.3 Breakaway Drawing Method

For long shafts or rods **with identical shapes** along their length **or changing in a regular pattern**, it is permissible to draw them in a **broken (discontinuous)** manner.

When dimensioning, still indicate the actual length.



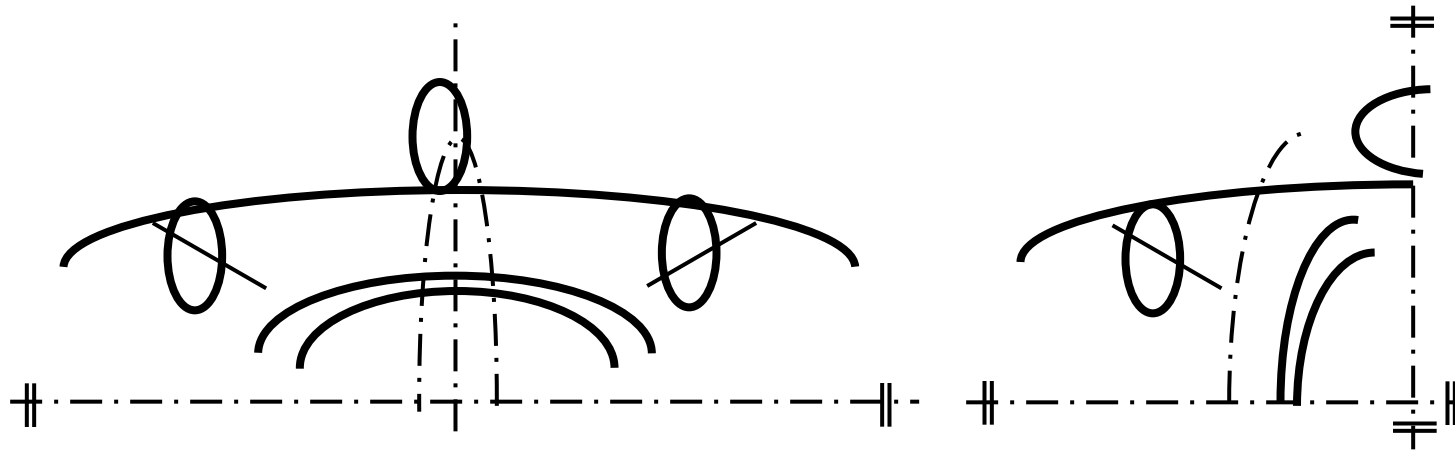
Breakaway drawing method for tension rod and sleeve.



Breakaway drawing method for stepped shaft.

6.4.4 Drawing Method for Symmetric Figures

When it does not cause confusion, **only half** or **a quarter** can be drawn. Two thin solid lines perpendicular to the centerline of symmetry should be drawn at both ends.



6.4.5 Other

When some parts of the machine are **too small** to be clearly shown, they can be drawn at **an enlarged scale locally**.

